# Fruit & Vegetable Department

## EDA Overview — Simplified Executive Brief

Foodland Wudinna, SA
Analysis Period: January – March 19 (2025 vs 2026)
Prepared: 23 March 2026

*This document summarises the full Exploratory Data Analysis into a simple, topic-based overview with key findings, analysis, and conclusions.*

# At a Glance

The full EDA covers 12 sections and 61 analysis cells. Below are the numbers that matter most.

| | | | |
|---|---|---|---|
| **-6.4%**<br>Revenue per Day YoY | **37.8%**<br>GP Margin (stable) | **+51.3%**<br>Avoidable Loss Growth | **$3,243**<br>Projected Annual Loss |
| **+505%**<br>REDUCED FV Scans Growth | **-21.2%**<br>Specials Cost (improving) | **$2,410**<br>Potential Labour Savings/yr | **$3,557**<br>Total Benefit if Actioned |

# Topic 1: Department Health

## What we found

Revenue per trading day dropped 6.4% compared to the same period in 2025. However, the gross profit margin actually improved slightly, from 37.3% to 37.8%. This means the store is not losing money on the items it sells — it is simply selling fewer units.

## What it means

The pricing model is healthy. The problem is volume loss, not margin erosion. Vegetables remain the largest revenue category. All sub-departments are down slightly but none show alarming declines.

> **Conclusion**
> The department's core pricing is sound. Focus should be on recovering volume (through better stock availability and competitive pricing on key items) rather than increasing margins.

# Topic 2: Waste & Markdowns

This is the most critical finding in the entire analysis.

## REDUCED Fruit & Veg (waste scanning)

- **2025:** 1.36 REDUCED scans per day, costing $1.09/day in lost GP.
- **2026:** 8.24 REDUCED scans per day, costing $5.12/day in lost GP.
- **Growth:** +505% in scan volume, +370% in GP loss.

This is the single biggest driver of the department's overall loss growth. The March 2026 rate reached 9.44 lines/day, suggesting the trend is accelerating.

### Barcoded markdowns (pre-packed items)

- **2025:** $9.29/day in true markdown losses.
- **2026:** $10.27/day — a +10.6% increase.

These are items where staff apply a markdown sticker and re-scan at a reduced price. After separating out Wednesday Specials (which are planned promotions), the true unplanned markdowns grew modestly.

### Wednesday Specials (planned promotions)

Good news: the cost of planned promotional pricing fell 21.2% year-on-year ($15.30 to $12.05/day). The store is running smarter, more targeted specials. No action needed here.

> **Conclusion**
> Waste (REDUCED FV) is the most urgent problem. It grew from $1.09 to $5.12 per day — a nearly 4x increase. This is caused by over-ordering or insufficient stock rotation, not by market conditions. Barcoded markdowns are a secondary concern. Specials are well-managed.

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## Topic 3: Pricing & Margin Traps

### Price elasticity

When prices go up on core items, customers buy less. The data shows a clear negative relationship between price increases and volume for the top 20 items. This is expected in a price-sensitive rural market like Wudinna.

### Margin Trap items

Some items had price increases in 2026 that backfired: the higher price pushed customers away, and the resulting volume drop meant the store earned fewer GP dollars overall — even though the per-unit margin was higher.

- **Green Grapes:** Biggest GP dollar loss from this effect.
- **Baby Lebanese Cucumbers:** Second highest impact.

> **Conclusion**
> Targeted price rollbacks on 3–5 Margin Trap items could recover lost GP. The goal is volume recovery, not margin-per-unit maximisation. Trial a 4-week rollback and measure the response.

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# Topic 4: Portfolio & Store Operations

## ABC classification

Items are classified into three tiers based on revenue contribution: A (top 80%), B (next 15%), C (bottom 5%). By comparing 2025 vs 2026 tiers, we identify:

- **Rising Stars:** Items that moved up to A-class in 2026 — these deserve more shelf space and higher order quantities.
- **Fallen Stars:** Items that dropped from A-class — consider reducing their shelf space.

## Basket Builders vs. Profit Drivers

Different items serve different roles. Staples like Bananas, Carrots, and Cucumbers drive foot traffic (Basket Builders). Higher-margin items like pre-packed salads generate the bulk of GP (Profit Drivers). Each group needs its own pricing strategy.

## Day-of-week rhythm

The store has a consistent weekly shopping pattern. This rhythm is stable year-on-year, which means ordering schedules and staffing can be reliably planned around it. Friday and Tuesday deliveries align well with demand peaks.

> **Conclusion**
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> The product portfolio has shifted slightly — some items gained importance, others declined. Ordering quantities should be reviewed against the updated ABC classification. The weekly rhythm is stable and reliable for scheduling.

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# Topic 5: Weather Impact

## Key findings

- **Temperature:** Hot days (>30°C) show lower average revenue. Extreme heat reduces foot traffic in this rural town where customers drive to shop.
- **Rainfall:** Rain days show lower sales compared to dry days, consistent with the rural driving hypothesis.
- **Category sensitivity:** Staples (Vegetables, Potatoes) are weather-resistant. Discretionary items (Berries, Herbs) show stronger weather effects.

## Practical use

Temperature adds marginal forecasting signal on top of day-of-week patterns. It is not a primary driver but can improve ordering accuracy for weather-sensitive categories on days with extreme heat forecasts.

> **Conclusion**

> Weather has a measurable but modest effect on sales. Day-of-week is the primary demand driver. Consider reducing orders for discretionary items when extreme heat (>35°C) or rain is forecast.

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# Topic 6: Data Quality Issues

Before building any forecasting model, six data quality issues were investigated:

1. **Open-ring scanning:** 2.1 lines/day have no product code. These cannot be attributed to specific items, which hides true category demand and degrades forecast accuracy.
2. **Negative GP entries:** Some items (mainly salad kits, mushrooms) are sold below cost. This may be distressed stock, incorrect cost entries, or promotional prices without matching cost updates.
3. **Item variant fragmentation:** Some products appear under 12+ separate PLU codes (e.g., Tomatoes). A forecast model sees 12 noisy low-volume series instead of one clean signal. These need consolidation.
4. **Small-volume items:** Chillies, Ginger, and Garlic sell <0.2 kg per scan. Forecasting in kilograms is unreliable — transaction lines are the more stable signal.
5. **Public holidays:** Six dates are confirmed store closures, not zero-demand days. These must be flagged in any model to avoid distorting baseline estimates.
6. **Wednesday Specials structure:** 58.7% of flagged low-GP events are planned promotions, not waste. Correctly separating these is essential for accurate loss measurement.

> **Conclusion**
> The data is usable but has known issues. The highest priority fixes are: eliminate open-ring scanning (assign PLUs to all items) and consolidate fragmented PLU groups before model training.

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# Topic 7: Forecasting Readiness

## Can we forecast demand?

Yes. Three validation tests were run:

1. **Baseline comparison:** A 4-week same-weekday moving average outperforms the naive same-week-prior benchmark for most items. Data-driven ordering adds value.
2. **Temperature signal:** Partial correlation confirms temperature adds some signal beyond day-of-week for certain items, but the effect is modest.
3. **Seasonal decomposition:** Most top-10 items have learnable weekly demand patterns with positive signal-to-noise ratios.

### Limitation

The dataset covers only January to mid-March (~78 trading days per year). All results reflect within-season performance only. A full-year model will need 12 months of continuous data to capture winter vs. summer patterns.

> **Conclusion**
> The data supports demand forecasting. A Prophet model with weekly seasonality, holiday flags, and Wednesday Special regressors is the recommended next step. Start with the top 10 items by revenue.

## Financial Impact Summary

All three loss categories combined (REDUCED FV, barcoded markdowns, negative GP) grew from $10.50/day to $15.89/day — a +51.3% increase year-on-year.

| Loss Category | 2025 $/day | 2026 $/day | Change |
|---|---|---|---|
| REDUCED FV (waste) | $1.09 | $5.12 | +370% |
| Barcoded markdowns | $9.29 | $10.27 | +10.6% |
| Negative GP entries | $0.12 | $0.49 | +308% |
| **TOTAL** | **$10.50** | **$15.89** | **+51.3%** |

- **Full-year central projection (2026):** $3,243 in avoidable losses.
- **Potential labour savings from demand forecasting:** ~$2,410/year.
- **Total recoverable benefit:** ~$3,557/year.

## What to Do — Priority Actions

Six actions, ranked by impact and urgency:

1. **Reduce waste scanning (NOW):** Audit the top 5 REDUCED FV items. Cut their order quantities by 15–20%. Target: below 5.0 REDUCED lines/day within 60 days.
2. **Eliminate open-ring scanning (NOW):** Assign a PLU to every item. Retrain cashiers. Target: below 0.5 open-ring lines/day.
3. **Price rollbacks on Margin Trap items (1–3 months):** Trial lower prices on Green Grapes and Baby Lebanese Cucumbers. Measure volume response over 4 weeks.

4. **Implement ordering aid (1–3 months):** Use a 4-week same-weekday moving average for top 20 items. Can be done in a spreadsheet — no new software needed.
5. **Deploy Prophet forecast model (3–12 months):** Build per-item forecasts with weekly seasonality, holidays, and specials flags. Target: MAPE <15% on top 10 items.
6. **Maintain waste log & monthly KPI review (ongoing):** Keep logging waste daily. Re-run analysis monthly to track progress against targets.

# Tools & Deliverables Built

This project has already produced the following:

- **Executive EDA Report (Jupyter Notebook):** 61-cell interactive analysis with 12 sections covering all topics above, professional visualisations, and embedded conclusions.
- **Financial Impact & KPI Report (.docx):** Detailed projection methodology, measured loss tables, full-year projections, and a KPI framework with targets.
- **Initiative Presentation (.pptx):** 12-slide management pitch covering findings, financial impact, tools built, and the ask.
- **This Overview (.docx):** Simplified, topic-based summary for quick reference.

*End of Overview*